

24EGMO1

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Problem

For a sequence $a_1 < a_2 < \dots < a_n$ of integers, a pair (a_i, a_j) with $1 \leq i < j \leq n$ is called *interesting* if there exists a pair (a_k, a_l) of integers with $1 \leq k < l \leq n$ such that

$$\frac{a_l - a_k}{a_j - a_i} = 2.$$

For each $n \geq 3$, find the largest possible number of interesting pairs in a sequence of length n .

Solution. The answer is $\binom{n-1}{2} + 1$.

We will show that at least $n - 2$ pairs are not *interesting*.

Let $m = \frac{a_1 + a_n}{2}$. For any $a_i \neq m$, we have 2 cases:

- $a_i > m$: In this case, the pair (a_1, a_i) is not *interesting* since $a_n - a_1 < 2(a_i - a_1)$.
- $a_i < m$: In this case, the pair (a_i, a_n) is not *interesting* since $a_n - a_1 < 2(a_n - a_i)$.

As a result, we must have at least $(n - 1) - 1$ pairs which are not *interesting*. (Note: Here (a_1, a_n) and (a_n, a_1) were both counted so we subtracted 1).

For construction, consider the following sequence: $a_1 = 0$, $a_i = 2^{i-2}$ for $i \geq 2$.
 $\square$